

PRAKASH REDDY PASHAM

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Software Developer with 2+ years of experience building **distributed backend systems and microservices** using **Java, Node.js, Go**, deployed on **AWS/ECS and Docker**. Skilled in **CI/CD, unit/integration testing, cloud-native deployments, monitoring (Prometheus and Grafana)**, and designing scalable, reliable, and maintainable systems.

EDUCATION

University at Buffalo, The State University of New York <i>Computer Science MPS DS GPA: 3.96/4.0</i> <i>Relevant Courses:</i> Machine Learning, Database Management System, Data Mining, AI, Deep Learning	Dec 2025 <i>Buffalo, NY</i>
Indian Institute of Technology Tirupati (IITTP) <i>Bachelor of Technology in Electrical Engineering GPA: 3.24/4.0</i>	Jul 2022 <i>Tirupati, India</i>

SKILLS

Languages & Frameworks	Java, Python, C++, C, JavaScript (ES6+), Spring Boot, Node.js, React, Postman API
Backend & Systems	Linux, Distributed Systems, REST API's, Microservices, OOP, Github, Jenkins, FastAPI
Databases & Cloud	MySQL, PostgreSQL, Cassandra, Redis, MongoDB, Kafka, AWS (EC2, S3), Docker
Observability & Tools	Prometheus, Grafana, OpenTelemetry, Datadog, JUnit, Mockito, Git
Security & Cryptography	AES-256, JWT, Secure Pipelines
Machine Learning & AI	PyTorch, TensorFlow, Scikit-learn, RAG, LLM's, Hugging Face, Generative AI, NLP
Frontend & UI	React, HTML5, CSS3, JavaScript, UI Integration, Streamlit
Methodologies	Agile, Scrum, Code Reviews, End-to-End Ownership, SDLC

EXPERIENCE

Software Engineer <i>PAYTM (One97 Communications Ltd)</i> - Developed and maintained large-scale backend microservices for credit and transaction workflows, supporting 14M+ users, including designing RESTful APIs for web applications. - Designed and implemented cloud-native, fault-tolerant distributed systems to ensure high availability, scalability, and data integrity. - Built and integrated object-oriented backend code with PostgreSQL, MySQL, and Cassandra to execute complex business logic efficiently. - Participated in peer code reviews and maintained high-quality, maintainable, and testable code using Git version control. Broke down larger projects into multiple phases, independently delivering medium-to-large features end-to-end, including rollout, monitoring, operational dashboards, KPIs, and on-call support. - Built reusable infrastructure components and performed deep Linux process, memory, and network debugging of client/server issues, improving system reliability by 25%. - Implemented production monitoring and alerting using Prometheus and Grafana, demonstrating operational excellence and reducing incident response time by 40%. - Automated deployment and scaling using Docker, Kubernetes, AWS, streamlining CI/CD pipelines and mentoring junior engineers on best practices.	Jun 2022 – Jun 2024 <i>Noida, India</i>
Research Assistant <i>Indian Institute of Technology Tirupati</i> - Addressed the challenge of accurate land-use classification in satellite imagery by developing a UNet++ deep learning model using TensorFlow and Keras. - Engineered a geospatial data preprocessing pipeline with Rasterio, converting multi-band GeoTIFF files into 16,000+ overlapping tiles for model training. - Improved feature learning and segmentation quality through dense skip connections and deep supervision in the encoder-decoder architecture. - Reduced the impact of class imbalance using LDAM, Dice, and IoU-based loss functions, achieving a 90.13% Dice score. - Built a production-ready end-to-end semantic segmentation pipeline covering data processing, model optimization, and inference for real-world satellite analysis.	Jan 2021 – Jun 2022 <i>Tirupati, India</i>

PROJECTS

AI-Enhanced Cybersecurity Threat Detection System (Full-Stack + RAG) Github - Developed a real-time binary Intrusion Detection System using Random Forest on the UNSW-NB15 dataset, achieving ROC-AUC: 0.91 and 87% accuracy after threshold tuning with risk-based classification (LOW/MEDIUM/HIGH). - Built a production-style backend using FastAPI + WebSockets to stream live threat alerts with 100 ms average inference latency, and persisted alerts in MySQL (Dockerized) for audit and historical analysis. - Integrated a RAG-based AI SOC Copilot grounded in curated cybersecurity runbooks, implementing similarity thresholding and refusal logic to prevent hallucinations while providing explainable incident response guidance.	Dec 2025 – Present
MediAid: Medical Document Processing & Retrieval System Github - Architected and implemented a Python-based RAG-driven LLM platform using embeddings, vector-based retrieval, and scalable backend services for semantic search over large document corpora. - Integrated vector search using OpenSearch-compatible indexing to support low-latency semantic retrieval in production environments. - Implemented embeddings and similarity-based retrieval using Sentence Transformers, integrated with PostgreSQL, Docker and monitored using Prometheus and Grafana.	Jan 2025 – Jun 2025
Age of Information in Networks - Developed a Python-based simulation for multi-component real-time IoT systems addressing data staleness. - Implemented metrics, monitoring, and error handling, achieving 15% reduction in data latency and 10% improvement in network efficiency.	Jul 2021 – Jul 2022